Newborough Warren

What's Happening to the Water Under the Dunes?

A condensed summary of a 21-year groundwater study

Hollingham, M. (2026). Hydrogeological Dynamics, Behavioural Clustering and Management Intervention Analysis at Newborough Warren Coastal Sand Dune Aquifer, Wales. Draft, 2026.
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The place

Newborough Warren sits at the southern tip of Anglesey — 1,300 hectares of sand dunes, wetlands and conifer plantation, designated as a Special Area of Conservation. Its dune slacks are seasonally flooded hollows between the dunes that support creeping willow, orchids and a rich community of wetland invertebrates. Their survival depends on groundwater sitting within 37 cm of the surface in summer; the difference between a thriving wet slack and an irreversibly degraded dry one falls within that margin. A 700-hectare Corsican pine plantation, planted between 1948 and 1965, intercepts roughly a quarter of incoming rainfall. This study monitored 88 dipwells monthly for 21 years (2005–2026), analysed with a 43-step data pipeline.

Five groundwater zones

A water balance model fitted to each well groups the site into five zones. C1 (Lake Edge) is shallow and fast-draining, closest to the ecological tipping point. C2 (Dune) shows moderate rainfall response. C3 (Western Residual) is a slow deep buffer. C4 (Main Forest) sits on thin sand over irregular bedrock: pine interception and a low-capacity substrate make it the most strongly amplifying zone in the network. C5 (Coastal Forest) is declining fastest of all zones. The thickness of sand beneath each area — not the tree cover — is the primary control on how severely evaporation draws down the water table in summer.

Zone	Character	Recharge sensitivity	Evaporation draw	Drainage rate	Canopy share
C1 Lake Edge	Shallow, fast-draining	4.58 (highest)	0.96 (lowest)	0.090 (fastest)	22%
C2 Dune	Open mature dune	3.98	1.77	0.066	26%
C3 W. Residual	Deep buffer	3.57	1.85	0.058	28%
C4 Main Forest	Pine on thin bedrock substrate	2.52 (lowest)	2.55 (highest)	0.020 (slowest)	40%
C5 Coastal Forest	Pine on deeper coastal sand	2.41	1.32	0.045	41%

Table 1. Summary of the five groundwater zones. Coefficients are dimensionless model parameters; larger values mean a stronger effect.

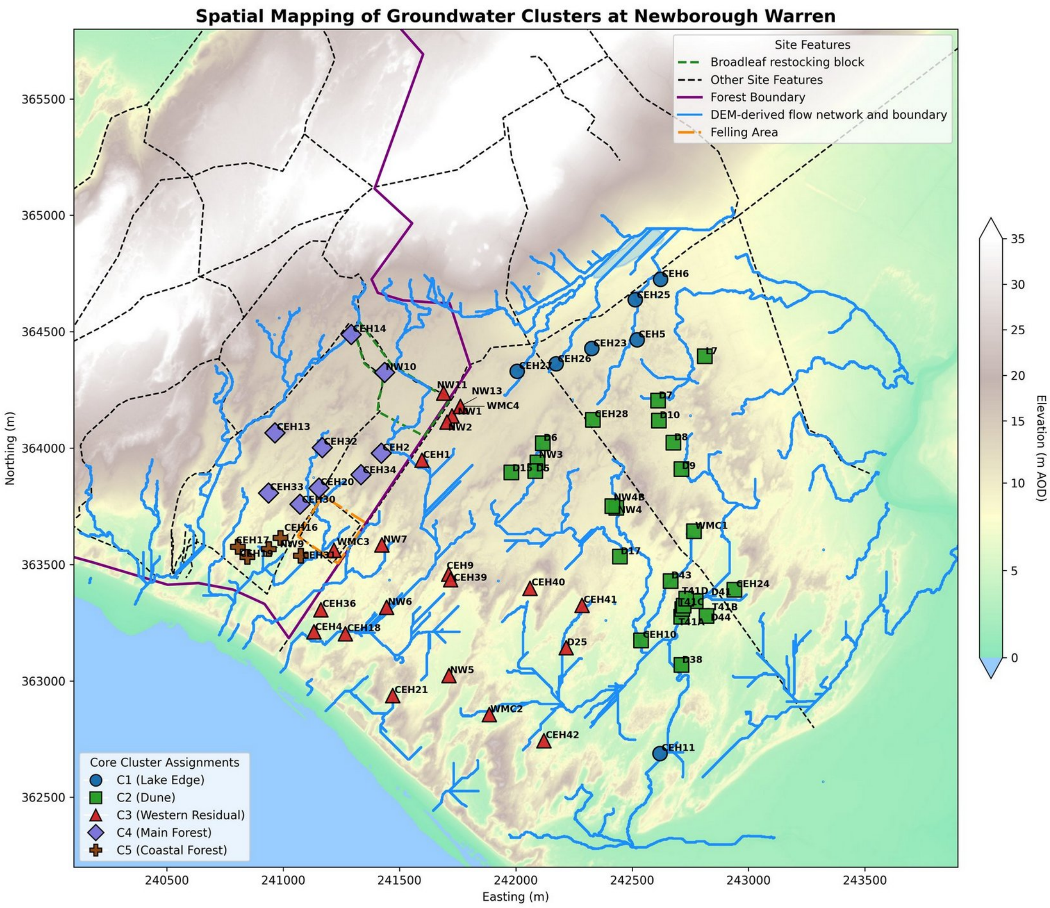

Figure 1. Spatial mapping of the five zones: C1 Lake Edge (blue), C2 Dune (green), C3 Western Residual (red), C4 Main Forest (purple), C5 Coastal Forest (brown).

What the data show

Summer is the critical season

The summer minimum water-table depth determines whether a slack floods the following winter. If it drops too low, no realistic volume of winter rainfall can recover it. Summer minima have been declining across all five zones. The Lake Edge zone is on course to cross the wet-slack threshold (−0.61 m, Curreli et al. 2013) as a long-term mean around 2030–2032; individual dry summers already reach it. UKCP18 projections indicate summer minima will deepen by a further 71–134 mm by the 2080s. The five-year mean spring baseline (MSL5) is more stable, projected to deepen by 21–39 mm over the same period — three to five times less than the summer minimum.

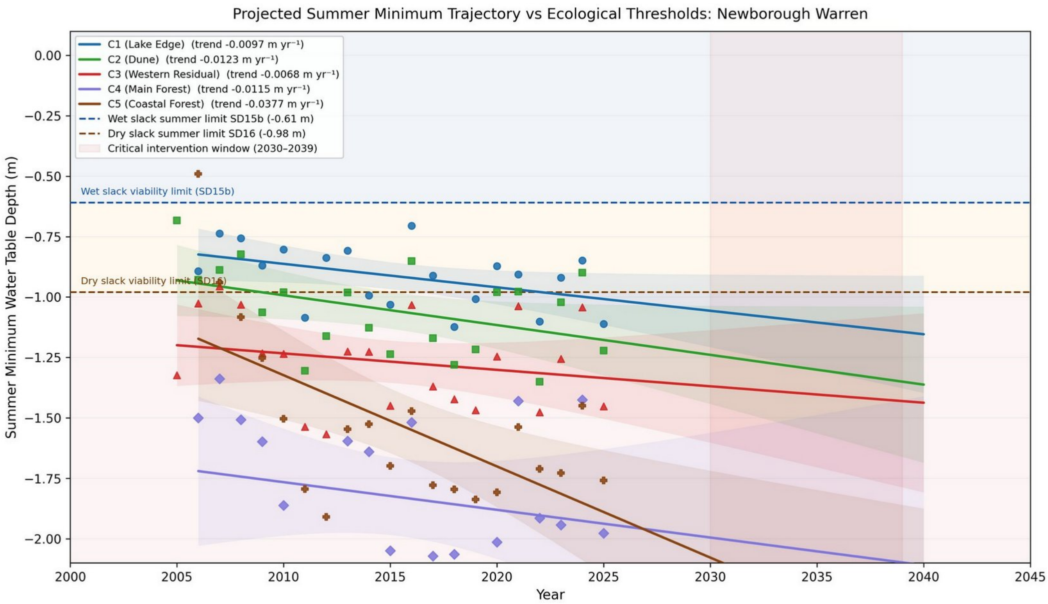

Figure 2. Projected summer minimum depth for all five zones. Wet-slack threshold −0.61 m (blue dashed); dry-slack threshold −0.98 m (brown dashed). Critical intervention window 2030–2039 shaded.

Dune scraping

CEH36 (scraped April 2015): three independent methods converge on a scraping benefit of 80–140 mm; the paired summer minimum shift against the unscraped control well was +195 mm ($p = 0.004$). The benefit is geometric — lowering the ground surface brings it permanently closer to the water table. The absolute water-table level at CEH36 has continued to decline since 2015; the improvement is maintained as a relative advantage over the unscraped condition. CEH18 and CEH21 were scraped in October 2023 at more seaward locations and have so far shown no detectable benefit in under two years of record; both sites are also subject to coastal-retreat boundary lowering that works against the intervention.

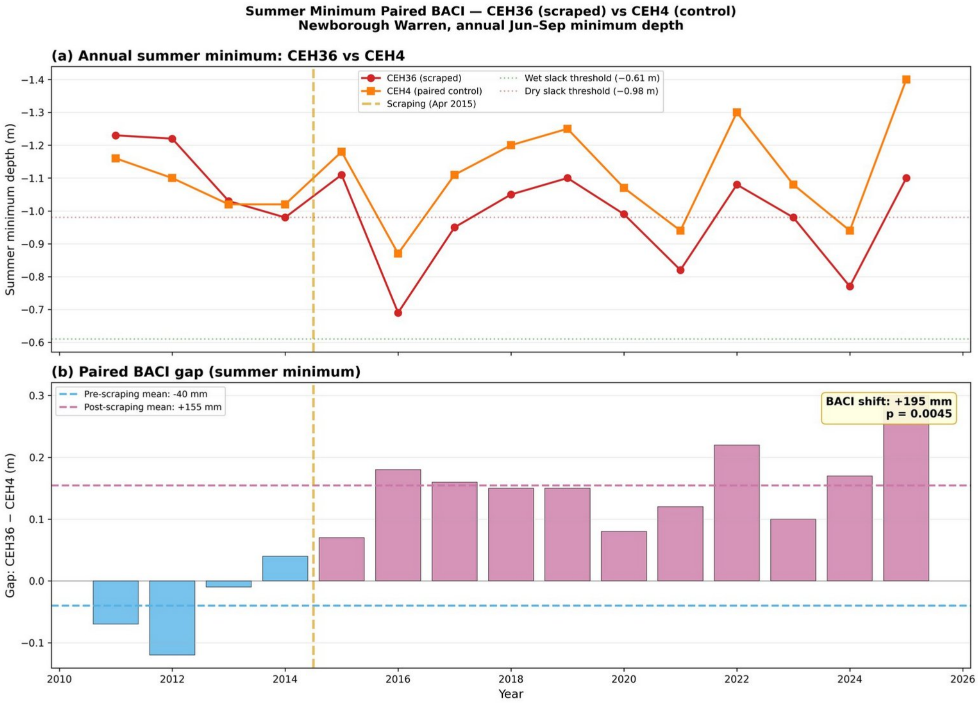

Figure 3. CEH36 (scraped, red) vs control CEH4 (orange). Paired BACI shift: +195 mm (p = 0.004). Top: annual summer minima. Bottom: gap between wells before and after scraping.

Tree felling

The December 2017 clearfell of 8.4 ha produced a statistically significant rise in mean monthly water levels at the core impact well of +120 mm relative to unfelled forest ($p < 0.001$). The forest edge showed +33 mm ($p = 0.19$, not significant). Summer minimum depths showed no corresponding improvement across any post-felling year. The clearfell raised mean monthly levels but produced no detectable benefit at the time of year that determines dune slack survival. The canopy acts as both a sink (intercepting rainfall) and a buffer (shielding the soil from direct summer evaporation); removing it recovers some winter recharge but increases summer drying by a comparable amount. Changes in the forest do not reach the open eastern dune slacks: the hydraulic divide routes forest groundwater south-westward toward Caernarfon Bay, not toward the Lake Edge and Dune zones.

A site-wide recharge decline

A decline in recharge efficiency — the proportion of rainfall that reaches the water table — operates across all five zones regardless of land cover or management history. It is consistent with changing rainfall patterns and represents a climate-driven trend. Coastal retreat is a separate driver: as the shoreline erodes, the aquifer loses volume at its seaward edge and the water table adjusts downward. Because groundwater moves slowly through sand, the declines visible today partly reflect erosion from several years ago. The coastal-retreat signal accounts for approximately 41% of the Coastal Forest zone's exceptional decline.

How water levels have changed

Comparing the pre-clearfell five-year window (springs 2013–2017) with the most recent comparable window (springs 2019–2023), the site-wide spring baseline deepened by an average of 97 mm. Of 59 wells with valid data in both windows, 56 deepened by more than 25 mm; none became shallower by more than 25 mm. The largest losses are at the south-western coastal margin; the eastern lake-edge area shows the smallest decline.

Looking at differential change since 2011, the Main Forest (C4) appears to be holding up relative to the network — but this reflects amplification of recent wet years (2021, 2024) rather than absolute improvement. The forest occupies the topographic and hydraulic high of the aquifer, furthest from any constant-head boundary, so its water table rises and falls more freely than other zones. The Lake Edge and Coastal Forest zones are declining relative to the network mean, driven by the retreating shoreline.

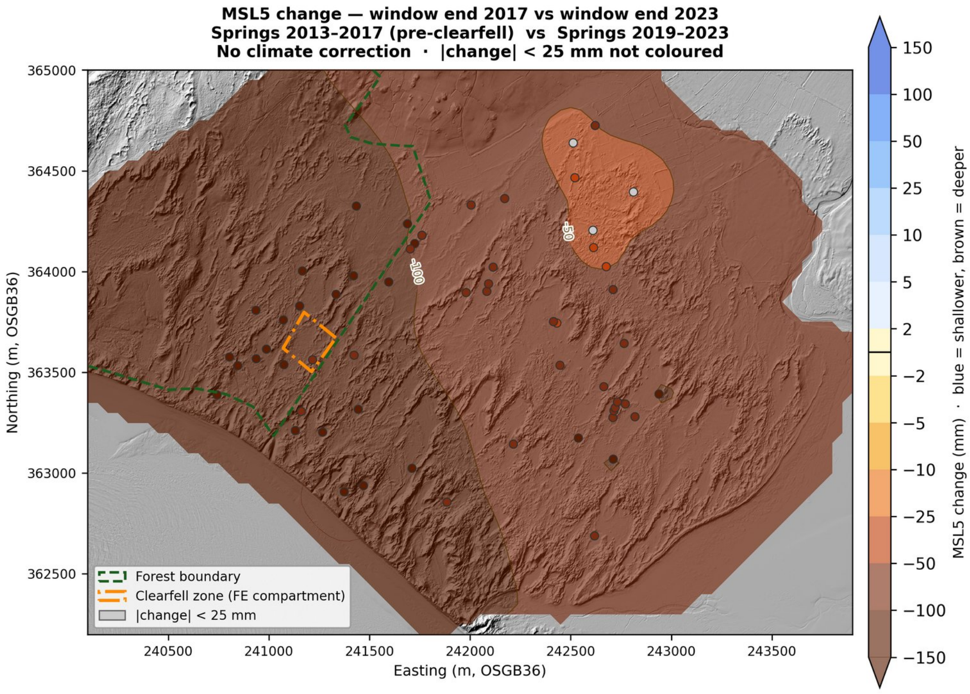

Figure 4. MSL5 change 2017→2023: 56 of 59 wells deepened >25 mm; none shallower. Largest losses at south-western coastal margin; clearfell zone (orange) indistinguishable from network trend.

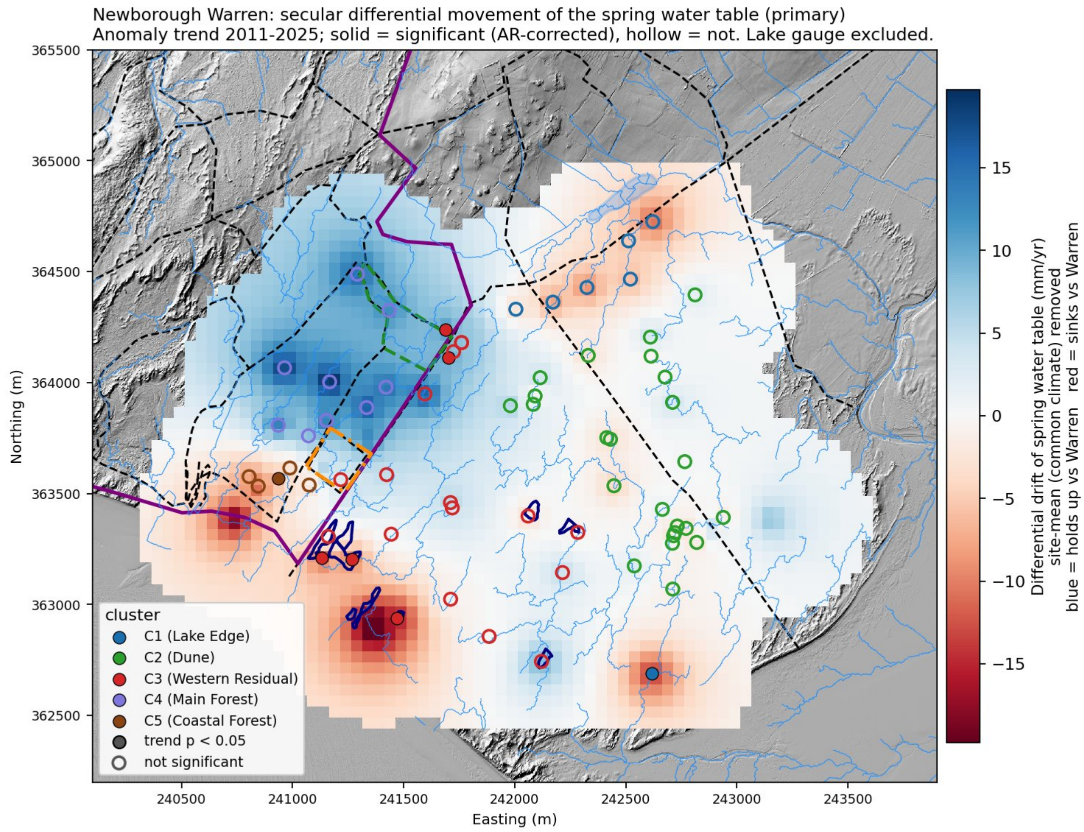

Figure 5. Differential spring movement 2011–2025. Blue = rising relative to network; red = sinking. Forest (C4) uniformly blue from amplification of recent wet springs. Lake Edge (C1) and Coastal Forest (C5) uniformly red from coastal retreat.

The scale of observed change in context

The management interventions studied to date have produced measurable effects at the local scale: the scraping benefit at CEH36 is statistically robust and ecologically significant, and the clearfell produced a detectable improvement in mean monthly water levels against forest controls. However, the site-wide spring baseline deepened by 97 mm between the 2017 and 2023 comparison windows — a change affecting 56 of 59 monitored wells simultaneously and driven by forces operating at the scale of the whole aquifer. Summer temperatures have trended upward at +0.014°C yr^{-1} since 1931, with a step increase of +0.94°C above baseline since 2013. The coastal-retreat signal accounts for approximately 41% of the Coastal Forest zone's exceptional decline and extends several hundred metres inland. The UKCP18 projections indicate a further summer minimum deepening of 71–134 mm by the 2080s. Added to the 97 mm already lost between the 2017 and 2023 comparison windows, cumulative losses from the pre-clearfell baseline are projected to reach 170–230 mm — comfortably exceeding any management effect observed in this record and operating simultaneously across the entire network rather than at a single location.

Interactive tools and further information

Three tools are freely available at newbroman.github.io/Newborough_Hydrology. The Scenario Viewer explores how UKCP18 climate projections and management options shift water levels across all five zones. The Flood Forecaster converts a current dipwell reading into a probability of winter flooding, updated through the season with live Met Office rainfall. The Seasonal Extremes Scatter plots every well against the Curreli et al. (2013) ecological viability thresholds. The full technical report, methods supplement and user manuals are available at the same address.